# Walter Zoff

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## Education

**Politecnico di Milano** Milan, Italy
*M.Sc. in Automation and Control Engineering — 110/110* *Sep. 2022 – Jul. 2025*

- Master's thesis: designed and trained a neural-network-based model (MLP) to simulate and optimize industrial process behavior, applying deep learning to a real engineering problem.
- Relevant coursework: Machine Learning, Optimization, Model Predictive Control, Statistical Methods, System Identification.

**Politecnico di Milano** Milan, Italy
*B.Sc. in Automation Engineering — 107/110* *Sep. 2019 – Jul. 2022*

## Experience

**Junior Consultant — Data & Analytics** Nov. 2025 – Present
*Target Reply S.r.l. (Reply Group) — Client: leading Italian bank* *Milan, Italy*

- Designing and developing interactive dashboards and reports in Looker / LookML on a banking-sector engagement, supporting business decision-making and KPI monitoring across multiple stakeholders.
- Building the analytical layer on BigQuery (GCP) through SQL data modeling and query optimization, feeding Looker views and explores.
- Manipulating and validating data in Python (Pandas) and supporting ETL / orchestration workflows across GCP and Databricks.
- Driving requirements gathering with business stakeholders, defining metrics, KPIs, and dashboard layouts in collaboration with the project team.

**Internal AI Initiatives — Target Reply** Nov. 2025 – Present
*Generative AI and BI migration projects (in parallel to client engagement)* *Milan, Italy*

- **MSTR-to-Qlik migration (end-to-end validated):** designed and developed an LLM orchestrator integrating Qlik MCP and Qlik APIs to automate the reconstruction of equivalent Qlik dashboards. Implemented parsing of MicroStrategy HTML dumps to extract semantic and structural metadata.
- **Change Impact Analyzer (Reply internal hackathon):** built a RAG-based application leveraging transformer models and LLM APIs with custom prompt engineering, designed to estimate Change Request effort and analyze data lineage on SQLite databases. Delivered the full stack including a React / Node.js frontend.

**Instrumentation & Automation Engineer** Jun. 2025 – Nov. 2025
*SIMECO S.p.A.* *Milan, Italy*

- Mechanized P&IDs and defined automation typicals (motors, control valves, on-off valves, safety devices) on industrial plant projects.
- Prepared instrument lists, I/O lists, and technical specifications for field instruments and control / safety valves; supported procurement through technical evaluation of supplier offers.
- Delivered detailed engineering deliverables (installation diagrams, cabling layouts, cable routing) and supported FAT/SAT and field activities.

## Selected Projects

**LLM Orchestrator for BI Platform Migration** | *Python, LLM APIs, MCP, Qlik APIs, HTML parsing* 2026

- End-to-end validated system that migrates MicroStrategy dashboards to Qlik Sense automatically, by orchestrating LLM calls against a Qlik MCP server and the Qlik REST APIs.
- Designed the parsing layer for MSTR HTML dumps and the prompt logic that drives dashboard reconstruction, reducing manual migration effort substantially.

**Change Impact Analyzer (Reply internal hackathon)** | *RAG, Transformers, LLM APIs, React, Node.js, SQLite* 2026

- Built a full-stack RAG application that estimates Change Request effort and analyzes data lineage across SQL databases, combining semantic retrieval with LLM-based reasoning.
- Engineered the prompt construction layer and integrated the React / Node.js frontend with the retrieval and inference pipeline.

**Neural-Network Optimization of Industrial Process (M.Sc. Thesis)** | *Python, PyTorch, Optimization* 2025

- Developed and trained an MLP to simulate the thermal behavior of an industrial process and optimize energy efficiency and uniformity, combining domain modeling with deep learning.
- Compared gradient-based methods and genetic algorithms for optimization in the trained network's input space.

## Technical Skills

**Languages**: Python, SQL, C/C++, MATLAB, JavaScript (basics)
**Data & ML**: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, MLPs, EDA, feature engineering, statistical modeling, optimization (gradient-based, genetic algorithms)
**Generative AI**: LLM APIs, Retrieval-Augmented Generation (RAG), prompt engineering, transformers, LLM orchestration, MCP (Model Context Protocol)
**Data Platforms & BI**: BigQuery, Databricks, Google Cloud Platform (GCP), Looker / LookML, Qlik Sense, MicroStrategy, ETL / data orchestration, data modeling
**Web & Tools**: React, Node.js, SQLite, Git, Excel (advanced, VBA)
**Languages (spoken)**: Italian (native), English (C1), Spanish (A2)

## Publications

W. Zoff et al., "Data-Driven Control of Thermoforming Machines," *IEEE Transactions on Control Systems Technology*, 2026. Co-authored a two-stage control strategy for thermoforming machines, integrating data-driven modeling with control design.

W. Zoff et al., "Data-Driven Modeling and Optimization of the Thermoforming Heating Phase," *IFAC-PapersOnLine* (IFAC Joint Conference on Computers, Cognition and Communication), September 2025. Co-authored a data-driven framework for the heating phase of thermoforming machines, addressing process modeling and optimization.

## Training & Certifications

**Google — Looker Skills Course:** data modeling and dashboard development in Looker, with focus on views, explores, and data visualization best practices.

**SAS — Hackathon Boot Camp (2026):** completed SAS's hands-on boot camp on tackling real-world business and industry challenges (in partnership with Intel and Microsoft).

**Politecnico di Milano — Identikit of Consultant: solve a case interview (Nov. 2024):** case interview methodology, problem structuring, and quantitative analysis in consulting contexts.

**Politecnico di Milano — MATLAB/Simulink for Control Systems Analysis and Design (Jan. 2022).**